

Jesus Andres Ruiz

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Education

University of Texas at San Antonio
Bachelor of Science in Mechanical Engineering

UTSA Honors College | **GPA: 3.6**
Expected May 2028

Skills

Software & Tools: SolidWorks, Fusion 360 (CAD + Simulation), FEA, MATLAB, Microsoft Office
Programming / Embedded: Arduino, Python, Git/GitHub
Manufacturing & Design: DFM, GD&T, Design Iteration, Rapid Prototyping, Failure/Root Cause Analysis
Fabrication & Tools: FDM & Continuous Carbon Fiber Printing, Wiring/Soldering, Mig/Tig Welding, CNC Machining
Materials & Testing: 6061-T6 Aluminum, Carbon Fiber Composites, Data Acquisition, Instrumentation, Schlieren Imaging

Relevant Experience

UTSA Intelligent Systems and Bionics Engineering Lab

San Antonio, TX

Undergraduate Research Assistant

May 2026 – Present

- Led fabrication of an exoskeleton hip module, working closely with graduate students to fabricate and assemble hardware across 3D printing and machining.
- Diagnosed and resolved 3D printer hardware/adhesion issues across 2 printer platforms (FDM and continuous carbon fiber), restoring consistent prototyping output.

UTSA Hypersonics Lab

San Antonio, TX

Undergraduate Research Volunteer

January 2026 – May 2026

- Fabricated precision diaphragms on the mill for a Mach 7.2 Ludwig tube, using material thickness and scored groove depth to set burst pressures between 200 and 400 psi across run conditions.
- Assisted researchers during hypersonic wind tunnel testing, helping prepare instrumentation, monitor runs, and collect experimental data from the wind tunnel.
- Set up and aligned laser-based Schlieren imaging systems to visualize shock waves and boundary-layer structures in hypersonic flow experiments.

UTSA Formula Society of Automotive Engineers

San Antonio, TX

Mechanical Engineer - Chassis / Brakes

August 2025 – Present

- Designed and machined aluminum caliper spacers for the 2025 competition car modeled in SolidWorks with GD&T thickness and parallelism callouts, then turned and faced them on the lathe to keep pad contact even and rotor clearance consistent across all four corners.
- Collaborated with a 5-person chassis and brakes sub-team to iterate designs under FSAE competition constraints, supporting fitment checks and assembly during vehicle build.

Project Experience

Exoskeleton Hip Module

San Antonio, TX

Undergraduate Research Assistant – Project Lead

May 2026 – Present

- Re-designed hip module components in Fusion 360 and validated redesigns using Fusion 360 static stress simulation following structural failure analysis of cracked cuffs, sliders, and struts on early prototypes.
- Manufactured parts via 3D printing (FDM, continuous carbon fiber), CNC machining, and outsourced cutting; corrected a 200-micron shoulder screw bore tolerance issue caused by anodizing buildup before final assembly.
- Iterated the hip abductor through 3 design/print revisions before finalizing it as a machined, anodized 6061-T6 aluminum part.

Bowden Cable Transmission – Ankle Actuation

San Antonio, TX

Undergraduate Research Assistant

May 2026 – Present

- Prototyped a Bowden cable transmission for remote ankle actuation, moving the actuator off the distal leg to reduce swing-leg inertia. Untested on subjects so far, but the design intent is to lower metabolic cost of walking.
- Designed, printed, and tested 2 TPU cable insert geometries (tapered vs. straight) to optimize routing and fit for cables.